

## Lecture 8 (10/2)

*Correlation (2.4), Estimators, Bias, and Consistency*

# Standardized variables

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Consider the standardized variables

$$Z_X = (X - \mu_X)/\sigma_X, \quad Z_Y = (Y - \mu_Y)/\sigma_Y$$

What is the regression coefficient of  $Z_Y$  on  $Z_X$ ?

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Therefore

$$\boxed{\beta_1 \sigma_X / \sigma_Y = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}}$$

This is called the correlation coefficient and denoted  $\rho$ .

# Coefficient of determination

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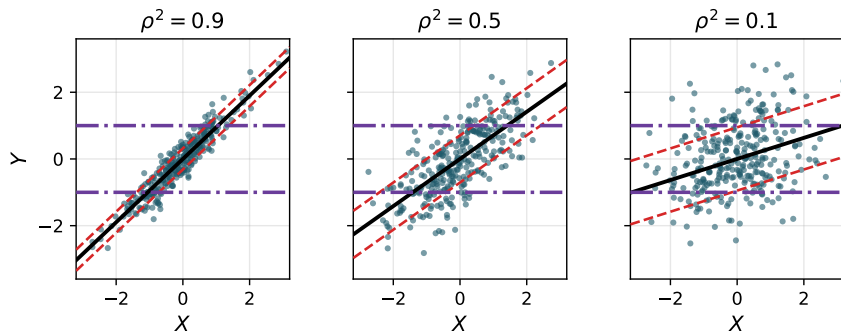
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Therefore

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$\rho^2$  is called the coefficient of determination

# Visualizing $\rho^2$



Here  $\mu_X = \mu_Y = 0$  and  $\sigma_X = \sigma_Y = 1$  (hence  $\text{cov}(X, Y) = \beta_1 = \rho$ ). The black lines show  $E[Y | X] = \rho X$  and red lines  $\pm \sigma_\epsilon = \pm \sqrt{1 - \rho^2}$ . Purple:  $\pm \sigma_Y = \pm 1$ .

## Summary: measures of association

|                    | Formula                                      | Units                      | Range               |
|--------------------|----------------------------------------------|----------------------------|---------------------|
| $\text{cov}(X, Y)$ |                                              | units $Y \times$ units $X$ | $(-\infty, \infty)$ |
| $\beta_1$          | $\frac{\text{cov}(X, Y)}{\sigma_X^2}$        | units $Y/\text{units } X$  | $(-\infty, \infty)$ |
| $\rho$             | $\frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$ | none                       | $[-1, 1]$           |
| $\rho^2$           | $1 - \frac{\sigma_\epsilon^2}{\sigma_Y^2}$   | none                       | $[0, 1]$            |

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- $\hat{\theta}$  is random – it will vary depending on which sample we collect.
- $\theta$  is a fixed, non-random number (in frequentist statistics, we will change this in unit 6).

# Sample distribution and standard error

## Example

Let  $Y_1, \dots, Y_N \sim \text{Normal}(\mu, \sigma^2)$  and estimate  $\mu$  with  $\hat{\mu} = \bar{Y} = \frac{1}{N} \sum_i Y_i$ .

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A sum of Normals is Normal, so we know the distribution of  $\hat{\mu}$  exactly:

$$\hat{\mu} \sim \text{Normal}\left(\mu, \frac{\sigma^2}{N}\right).$$

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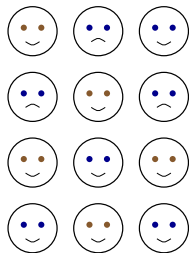
$$\hat{\mu} \sim \text{Normal}\left(\mu, \frac{\sigma^2}{N}\right).$$

The sample distribution is the distribution of  $\hat{\mu}$  over replicates of the data. The standard deviation of the sample distribution is the standard error

$$\text{se}(\hat{\mu}) = \frac{\sigma}{\sqrt{N}}.$$

# Replicates

Back to our population, where a fraction  $q = 1/3$  of people are sad. Each time we collect data we get a new sample of  $N = 12$  people – a replicate – and a new estimate  $\hat{q}$ :

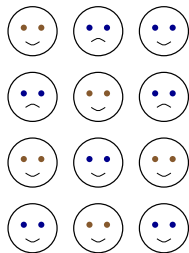


Replicate 1

$$\hat{q} = 3/12$$

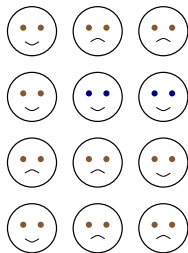
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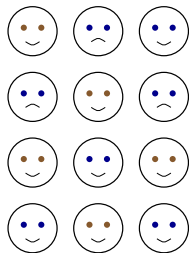


Replicate 2

$$\hat{q} = 6/12$$

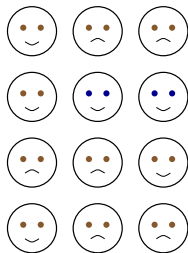
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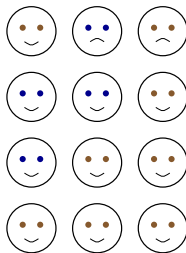
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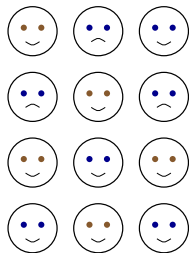


Replicate 3

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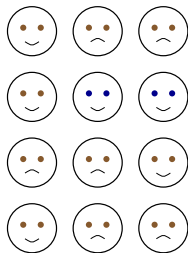
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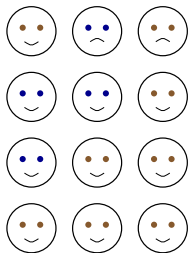
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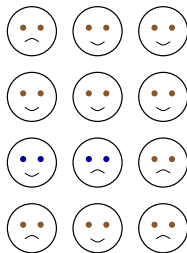
Replicate 2

$$\hat{q} = 6/12$$



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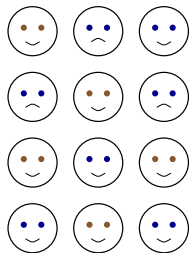


Replicate 4

$$\hat{q} = 5/12$$

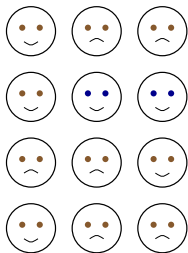
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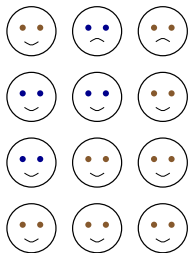
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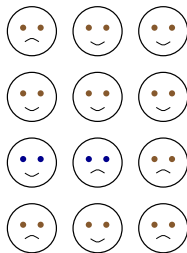
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Replicate 3

$$\hat{q} = 2/12$$



Replicate 4

$$\hat{q} = 5/12$$

$\hat{q}$  changes from replicate to replicate, while  $q$  stays fixed. The distribution of  $\hat{q}$  over (infinitely many) replicates is its sample distribution; its spread is  $se(\hat{q})$ .

# Estimating $q$

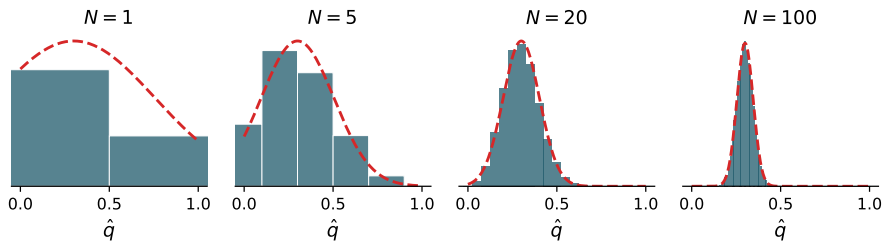
## Example

Let  $Y_1, \dots, Y_N \sim \text{Bernoulli}(q)$  and estimate  $q$  with  $\hat{q} = \overline{Y}$ .

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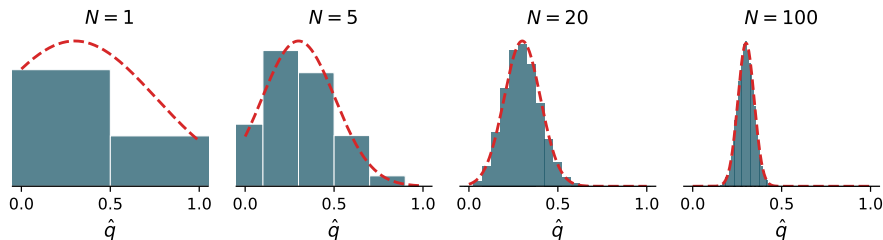
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$\hat{q}$  is not Normal, but by the CLT the Normal result carries over for large  $N$ , with  $\sigma^2 = q(1 - q)$ :

$$\hat{q} \approx \text{Normal} \left( q, \frac{q(1 - q)}{N} \right), \quad \text{se}(\hat{q}) = \sqrt{\frac{q(1 - q)}{N}}.$$

# Bias and consistency

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$\hat{\theta}$  is consistent if

$$E[\hat{\theta}] \rightarrow \theta \quad \text{and} \quad \text{se}(\hat{\theta}) \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

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### Example

$$\hat{\sigma}_0^2 = \frac{1}{n} \sum_i (Y_i - \bar{Y})^2$$

Is this unbiased?

## Bessel's correction

Add and subtract  $\mu$ , expand, and use  $\sum_i (Y_i - \mu) = n(\bar{Y} - \mu)$ :

$$\sum_i (Y_i - \bar{Y})^2 = \sum_i [(Y_i - \mu) - (\bar{Y} - \mu)]^2$$

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Taking expectations,  $E[(Y_i - \mu)^2] = \sigma^2$  and  $E[(\bar{Y} - \mu)^2] = \text{var}(\bar{Y}) = \sigma^2/n$ :

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So  $E[\hat{\sigma}_0^2] = \frac{n-1}{n}\sigma^2$  is biased and consistent. Dividing by  $n-1$  (Bessel's correction) gives a unbiased estimator

$$\hat{\sigma}^2 = \frac{1}{n-1} \sum_i (Y_i - \bar{Y})^2, \quad E[\hat{\sigma}^2] = \sigma^2$$